Janit Rajkarnikar

+1 (601) 913-1805 · rajkarnikarjanit22@gmail.com · anijani.github.io · github.com/aniJani

Research Interests

Machine learning, representation learning, and information access (retrieval, ranking, and synthesis). Interested in privacy, interpretability, and building deployable systems that support trustworthy decision-making. Open to exploring new directions across ML and information science.

Education

University of Southern Mississippi *May 2026*

Bachelor of Science in Computer Science, *Minor in Mathematics and Data Analysis* *Hattiesburg, MS*

GPA: 4.0

Coursework: Algorithms, Machine Learning, Data Structures, Operating Systems, Real Analysis, Linear Algebra, Discrete Mathematics, Probability & Statistics, Database Systems, Programming Languages

Honors & Awards

President's List (All Semesters), Academic Excellence Scholarship, Runner-up Voxo Hackathon at USM

Publications

[1] **J. Rajkarnikar**, N. Poudel, and N. Rahimi, "Unsupervised Anomaly Detection in OpenStack Logs via Fine-Tuned RoBERTa Embeddings," *Journal of Cybersecurity, Digital Forensics and Jurisprudence*, vol. 1, pp. 9–21, 2025. [Paper]

[2] D. Precious-Esue, **J. Rajkarnikar**, B. Bellrose, et al., "Ensemble Machine Learning Approach to Phishing Website Detection," in *Proc. 40th International Conference on Computers and Their Applications (CATA 2025)*, San Francisco, CA, USA, Mar. 2025, pp. 109–121. Springer CCIS, vol. 2435. [DOI]

Research Experience

Advised by Dr. Zhaoxian Zhou *Sep 2025 – Present*

- Developed **CAMMF**, a cross-attention framework for integrating heterogeneous time-series information from macroeconomic indicators and market signals for return prediction
- Designed alignment and leakage controls for multi-source temporal data; improved AUROC by 8.9% over baselines
- Built interpretability analyses that attribute predictions to specific information sources/lags across regimes (attention attribution + **SHAP**)

Advised by Dr. Nick Rahimi *May 2024 – Present*

- Led representation learning pipeline for unstructured OpenStack logs using **RoBERTa** embeddings with **LoRA/PEFT**; achieved **0.97 F1** and **0.99 ROC-AUC** with Isolation Forest *(Publication [1])*
- Conducted empirical study of ensemble learning for phishing website detection; Bagging achieved **98.66% accuracy** *(Publication [2])*

Projects

RepoSynth – Repository Understanding & Retrieval Platform (reposynth.com) ***Python, Rust, TypeScript***

- Built pipeline for extracting semantic representations and structural signals (AST/dependency graphs) to support repository understanding and retrieval (**SentenceTransformers + FAISS**)
- Implemented **Rust Tree-sitter** parsing daemon and multi-stage caching, reducing re-analysis time by 95%

Emoki – Emotion Inference System ***React Native, Android, Flask, ONNX***

- Deployed quantized **RoBERTa (ONNX)** for real-time affect inference; engineered end-to-end client/server pipeline

- Built Android Accessibility-based text capture and on-device processing workflow, highlighting system design tradeoffs between inference utility and user privacy

Suga – Workflow Automation Companion (suga.cx) ***FastAPI, Next.js, Supabase, Electron***

- Built an automation capture-and-replay system that converts observed user actions into reusable workflows

Experience

Software Intern *Sep 2025 – Present*
CloudCrust LLC *Hattiesburg, MS*

- Built and shipped production web/mobile features (**Next.js, React Native**) including geospatial visualization and location-based data workflows
- Implemented authentication and subscription flows for deployed applications

Software Engineer Intern *Aug 2024 – May 2025*
Woafmeow, Inc. *Remote*

- Developed backend features in **FastAPI/PostgreSQL** supporting secure multi-tenant workflows

Technical Skills

ML/Data: PyTorch, SentenceTransformers, FAISS, SHAP, ONNX, SQL, PostgreSQL, Redis, MongoDB
Systems/Backend: FastAPI, Flask, Node.js/Express, Docker, AWS, Azure, Supabase, Git
Languages: Python, TypeScript, C/C++, C#
Frontend/Mobile: React, Next.js, React Native, Expo, Electron

Service & Leadership

Vice Head of SWE, Google Developer Group – Coordinated technical workshops and campus engineering events
President, Nepalese Student Association – Led 15+ volunteers; organized 250+ attendee cultural/academic events
Treasurer, Cinema Appreciation Club – Managed finances and coordinated screenings/events